HETSA POWERSUITE

The Engineering Field Platform - Southern Africa.

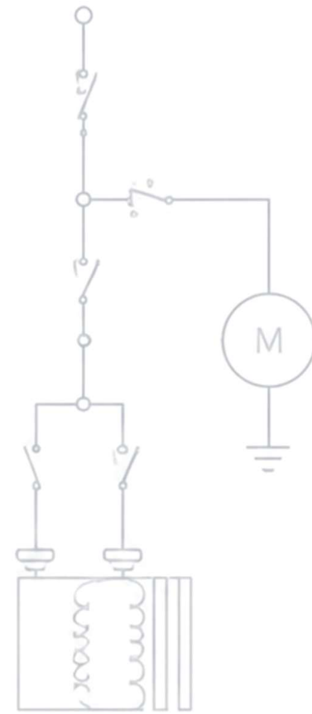
USER MANUAL

Version 1.0 (First Edition)

DOWNLOAD THE APP

hetsak-sys.github.io/
hetsa-powersuite.apk

Contents



1. Introduction

PowerSuite is an offline engineering field platform for a wide range of electrical professionals — including engineers, technicians, electricians, and artisans — working across Southern Africa. It seeks to replace or minimise manual calculations, reference books, and error-prone spreadsheets with a single tool designed for real field conditions, taking into consideration high-altitude impact on equipment efficiency, unreliable connectivity, and rugged daily use.

Every core calculation works fully offline. Only license verification requires an internet connection, and only briefly.

PowerSuite's standards references (SANS, NRS, MHSA, the OHS Act) are South African instruments specifically. If you're working elsewhere in Southern Africa, it's worth confirming the equivalent standard or regulation for your country before relying on a South African-specific figure or compliance check.

Who this manual is for

Field electrical professionals using PowerSuite day-to-day. It assumes basic familiarity with the electrical concepts involved (voltage, current, power factor, earthing, protection) but not with the app itself.

Who PowerSuite is for

- Electrical technicians and engineers
- Contractors and consulting engineers
- Mine electrical staff
- Utility and municipal electrical personnel
- Solar / renewable installers
- Training colleges and students
- Inspectors and compliance assessors

What PowerSuite Includes

This is the first edition of PowerSuite. Everything below is included from day one — there is no prior version this is being compared against.

- Motors & Drives — Full Load Current, DOL/star-delta starting current, contactor & overload relay selection (IEC 60947-4-1), breaker sizing, motor protection and core-balance relay settings, voltage-dip estimation, and IE-class efficiency comparison.
- Cable & Wiring — conductor sizing (IEC 60364 ampacity tables with derating), detailed voltage drop, fault current/withstand checks, trailing cable sizing, conduit fill, gland selection, and underground reticulation (direct-buried sizing, duct derating, route fault level).
- Earthing — electrode resistance (Dwight's formula), touch/step voltage, conductor sizing, and fault-loop impedance, referenced to IEC 62305 / SANS 10199.
- Protection — arc flash screening estimate, transformer differential protection, protection grading, a full time-current coordination chain builder with a log-log TCC plot, earth fault (core-balance/HR-SEF) sizing, and relay/CT selection guidance.
- Power Systems — transformer, power-factor correction, generator sizing (a full multi-load flow: loads → generator → transformer → fault level), busbar sizing, and motor starting studies.
- Power Quality — harmonics (THD and K-Factor per IEC 61000 / IEEE 519), battery/UPS sizing, and lighting calculations.
- Renewable Energy — PV array sizing, battery sizing for off-grid and hybrid systems, NRS 097-2 grid-tie compliance, and hybrid (solar + battery + generator) system design.
- Installation Design — load assessment, distribution board sizing, final-circuit design, and area/exterior lighting.

- Overhead Reticulation — bare conductor sizing, pole spacing, pole planting, clearances (including HV and EHV transmission voltages), fittings and structures, a full construction reference with an interactive pre-energization checklist, and a fault-finding/maintenance reference.
- Unit Converter — 12 engineering categories.
- Formula Reference — a searchable library of the formulas used throughout the app, plus your own saved formulas.
- Quick Math — a standard calculator alongside a formula-programming tool for repeat site calculations.
- PDF export, calculation History, and per-site Settings (voltage, frequency, altitude, currency) across every module.

2. Getting Started

2.1 Installing the app

PowerSuite is distributed as an Android APK file, not through the Google Play Store. You'll receive the APK on the download link: hetsak-sys.github.io/hetsa-powersuite/hetsa-powersuite.apk. On most Android phones you'll need to allow "install from unknown sources" the first time — Android will prompt you for this automatically when you open the file.

Scan to download the APK directly.

2.2 First launch and the trial

The first time you open PowerSuite, it registers a 90-day trial with the license server. This requires an internet connection for that first launch only. Once registered, the trial clock is tracked on the server — not on your phone — so clearing app data or reinstalling does not reset it.

Note: If PowerSuite shows "Couldn't reach the license server" on first launch, check your internet connection and tap Retry. The license server can take up to a minute to respond if it hasn't been used recently — this is normal, not a fault.

2.3 Site Profile

Before your first calculation, it's worth setting your Site Profile in Settings — this pre-fills sensible defaults (voltage, power factor, motor efficiency, ambient temperature, altitude) across the calculators based on where you're working, saving repeated typing.

3. Home Screen

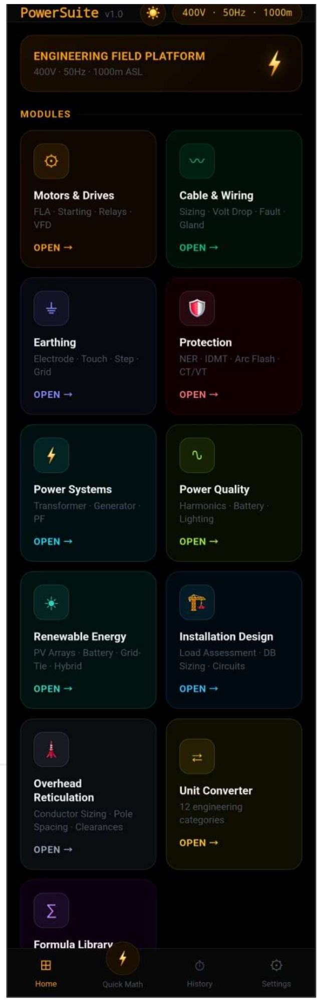


Figure — Home screen: site summary and module grid

The Home screen shows your current site summary (voltage, frequency, altitude) at the top, followed by a grid of modules. Tap any module to open it. Most modules contain several sub-tabs across the top of the screen — swipe or tap to switch between them.

Every module on this screen runs fully offline — the only feature on the entire phone that ever needs a connection is the license check itself, and only briefly.

4. Typical Workflows

The chapters that follow document each module tab by tab. This chapter shows how three common field jobs actually move through the app end to end, so you can see the shape of a real workflow before diving into the module reference.

Workflow A — Sizing a motor feeder

Motors & Drives → FLA, to get the motor's Full Load Current. This carries automatically into Cable & Wiring — Sizing, to recommend a conductor. Run Cable & Wiring — Volt Drop to confirm the cable meets your voltage-drop limit for the actual run length, since a cable that only just passes on current alone can still fail on volt drop over a long run. Only once the cable is confirmed viable, move to Protection to size the breaker and any protection settings — protection follows the cable that will actually be installed, not the other way around. Finish with Export PDF for a record of the feeder.

Workflow B — Generator or standby power design

Power Systems → Generator, using Load Schedule Sizing mode (a mode inside the Generator tab itself, not a separate screen). Work through its four-stage flow — Loads, Gen, Transformer, Fault Level — in order. Take the resulting fault level to Protection → Coord. Study to check that your protection devices grade correctly against it. Finish with Export PDF — the combined Gen/Transformer/Fault-Level report is reachable from any of the three sizing stages, so you don't need three separate exports.

Workflow C — Solar installation

Solar jobs split into two different paths depending on the system type — see "How the Four Tabs Connect" below for why these paths aren't identical.

C1 — Off-grid or hybrid system: Renewable Energy → Array, which leads with a load-only sizing result (Required Wp) before any panel datasheet is needed. Then Battery, which also leads with a load-only result (Required Ah) — its charge-controller section can optionally pull Array's current/voltage as a starting point if you want to verify a specific product pairing, but the core battery sizing doesn't need Array at all. If the system includes a generator for extended low-sun periods, go to Hybrid, which can pull in a generator sizing result directly from Power Systems → Generator if you've already sized one for this site. Finish with Export PDF.

C2 — Grid-tied system: Renewable Energy → Array, to get your load-only required Wp (Grid-Tie's recommended inverter AC rating is also surfaced directly on this same screen, since it's derived from the same Wp figure with no extra input needed). Then Grid-Tie for the NRS 097-2 SSEG category, DC:AC ratio check, and compliance checklist. Finish with Export PDF.

How the Four Tabs Connect

Renewable Energy's four sub-tabs don't all depend on each other the same way. Knowing the real dependency map helps you understand why Workflow C splits into two paths above, and why some fields prefill while others don't:

- **Array** is fully independent — it sizes from your daily energy target alone and needs nothing from the other three tabs.
- **Battery's** core bank-sizing math is independent of Array too — it works from your load and autonomy requirement alone. Only its charge-controller section optionally prefills from Array's current/voltage, and only if you choose to verify a specific product pairing.
- **Grid-Tie** genuinely needs Array's Wp figure — the DC:AC ratio it checks can't be calculated without both the array's power and the inverter's rating.
- **Hybrid** needs its own low-sun-day production figure, plus a generator sizing result carried over from Power Systems if you have one — but it never uses Array's numbers at all, even though it sits in the same module.

Note: These three workflows are the common cases, not the only paths through the app. Every module can also be used on its own for a single, isolated calculation — you don't need to follow a full workflow if you only need one figure.

5. Motors & Drives

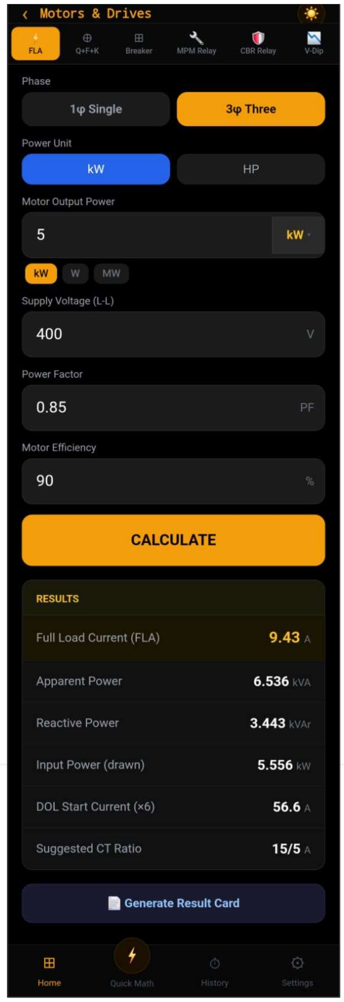


Figure — Motors & Drives: FLA calculation result

Covers everything needed to select and protect a motor starter, from full-load current through to protection relay coordination.

Sub-tab	What it does
FLA	Calculates Full Load Current from motor power (kW or HP), voltage, power factor, and efficiency. Also shows apparent/reactive power, DOL starting current, and a suggested CT ratio.
Q+F+K	Selects the main contactor (Q), thermal overload relay (F), and control relay (K) per IEC 60947-4-1, cross-referenced against Schneider, Eaton, Siemens, and Allen-Bradley part numbers.
Breaker	MCCB sizing for the motor circuit.
MPM Relay	Motor Protection Monitor relay settings.
CBR Relay	Core Balance Relay (earth fault protection) sizing.
V-Dip	Estimates the voltage dip caused by motor starting, based on transformer kVA and impedance — useful for checking whether DOL starting is acceptable on a given supply.
IE Class	Compares IE1–IE4 motor efficiency classes.

Note: The FLA you calculate on the FLA tab automatically carries into Cable & Wiring's Sizing, Volt Drop, Trailing Cable, and VFD tabs — you don't need to re-type it. A green "FLA loaded from Motor tab" banner confirms this and shows the values it pulled through; you can still overwrite the field if you're sizing for a different scenario.

6. Cable & Wiring

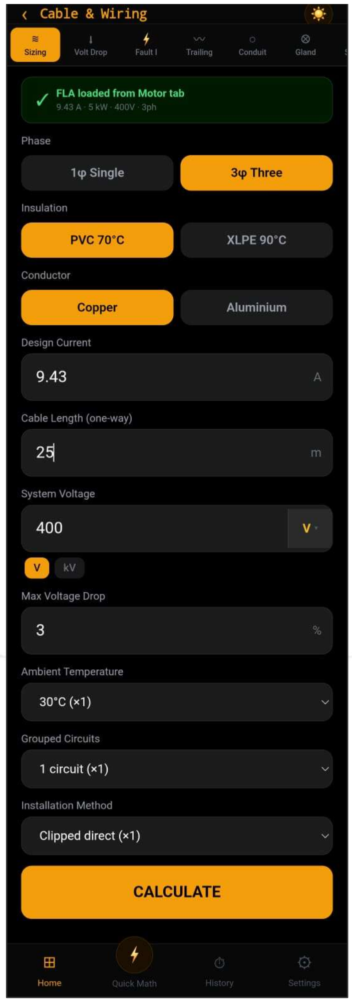


Figure — Cable & Wiring: Sizing result with full CSA table

Cable sizing, derating, and fault-current checks for fixed, flexible/trailing, and underground installations.

Sub-tab	What it does
Sizing	Recommends a conductor size from IEC 60364 ampacity tables, applying derating for ambient temperature, grouping, and installation method, and checking voltage drop against your limit.
Volt Drop	Detailed IEC-method voltage drop calculation (R·cosφ + X·sinφ), more precise than the simple resistivity method used in Sizing.
Fault I	3-phase and 1-phase fault current at the end of a cable run.
Trailing	Sizing for mining trailing (flexible) cables, with a fixed 0.85 derating factor for flexible use.
Conduit	Conduit fill percentage for a given set of conductors.
Gland	Cable gland selection based on cable outer diameter.
Schedule	A running cable schedule you can build up across multiple circuits, exportable as CSV.
VFD	Screened VFD output cable sizing, with harmonic and screening derating applied, and a check against the typical 50 m unfiltered length limit.

Sub-tab	What it does
Buried	Direct-buried underground cable sizing — soil thermal resistivity (or a qualitative soil-nature setting), ground temperature, and grouped-circuit derating applied together, per IEC 60364-5-52.
Duct	Duct-bank underground cable derating — uses the duct-specific base ampacity table, which runs lower than the direct-buried table at the same current, so expect a larger recommended size for the same load in a duct run.
Route Fault	Fault current at each point along a multi-segment underground route, tracking how fault current drops as impedance accumulates through successive cable sections.

Note: Duct Derating currently reuses the direct-buried grouping-derating table, since a duct-bank-specific grouping table could not be verified against IEC 60364-5-52 with confidence at the time of writing. This is flagged directly on the Duct tab — treat it as a reasonable conservative estimate for typical field-quick sizing, not a substitute for a full thermal study on a large duct-bank installation.

7. Earthing

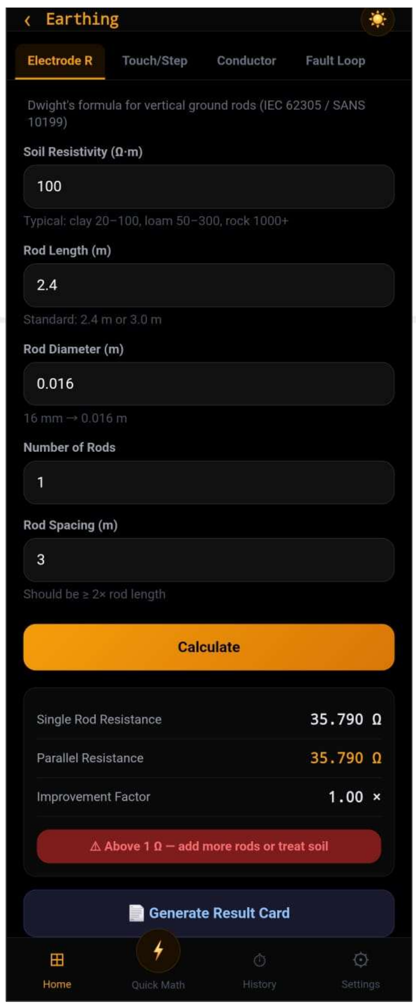


Figure — Earthing: Electrode Resistance (Dwight's formula)

Earthing system design per IEEE 80 and related standards.

Sub-tab	What it does
Electrode R	Earth electrode resistance using the Dwight formula.
Touch/Step	Touch and step voltage limits, and comparison against calculated values.
Conductor	Earth conductor sizing via the adiabatic equation.

Sub-tab	What it does
Fault Loop	Earth fault loop impedance.

8. Protection

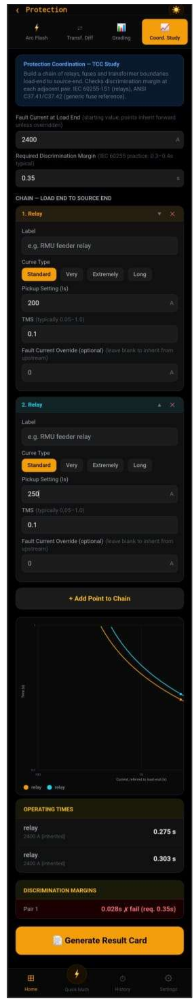


Figure — Protection: Coord. Study TCC plot

Relay coordination, CT checks, insulation testing, arc flash estimation, transformer differential protection, earth fault protection, relay function selection, and full time-current coordination studies.

Sub-tab	What it does
NER Size	Neutral Earthing Resistor sizing — sets the continuous current rating and earth fault current limit.
NCRT	Neutral Current/Ratio Transformer monitoring settings.
Earth Fault	Earth fault current and pickup sizing for both High-Resistance (HR) and solidly-earthed systems — covers HR-SEF (sensitive earth fault) pickup with an explicit warning if the setting would never operate or is set too sensitively for CT measurement error, and solid-earth pickup with a nuisance-trip warning if the time delay is set too short.
Relay Select	Recommends which protection relay functions apply for your application (feeder, transformer, or generator) and earthing method (HR or solidly-earthed), plus the minimum standard CT accuracy class (5P/10P) needed for the fault level. Generator applications are explicitly flagged as out of this tab's scope rather than given a guessed recommendation.

Sub-tab	What it does
IDMT	IDMT relay operating time per IEC 60255-151, across Standard/Very/Extremely Inverse and Long Time Inverse curves. Shows plug setting multiple and operating time, plus a note on typical grading margins between relay tiers.
CT Burden	Checks that cable wiring plus relay burden stays within a CT's rated burden — accounts for cable run length and cross-section as well as the connected relay's own burden.
PI/DAR	Insulation resistance testing per IEEE 43 — calculates Dielectric Absorption Ratio and Polarization Index from your 30-second, 1-minute, and (optional) 10-minute readings, with the standard's general guidance bands for each.
Arc Flash	A simplified LV (below 1kV) incident-energy estimate using the IEEE 1584 empirical method. Clearly marked as an estimate only — not a substitute for a full arc-flash study — since PPE selection should never rest on this figure alone.
Transf. Diff	Transformer differential protection: CT ratio matching either side of the transformer, the resulting CT secondary mismatch, and typical starting points for minimum pickup and slope settings.
Grading	A reference table of typical discrimination margins between protection device tiers (fuse-fuse, relay-relay, relay-fuse, breaker-breaker) — general industry starting points, not a substitute for a full coordination study.
Coord. Study	A full protection coordination (time-current) study builder — add a chain of protection points (relays, fuses, transformers) in sequence, and see each device's operating time plotted together on a log-log time-current curve, with a check on whether adjacent devices maintain adequate grading margin at the fault level you specify.

Note: The Arc Flash tab is explicitly a screening estimate, not an engineering study. IEEE 1584 is used because there is no distinct IEC arc-flash energy calculation standard — it's the method used internationally, including in IEC-based markets. Always have a qualified person perform a full arc-flash study before making PPE or safety decisions.

Note: The PI/DAR and Arc Flash tabs feed real safety go/no-go decisions — always cross-check against a qualified assessment before relying on either for a live decision.

9. Power Systems

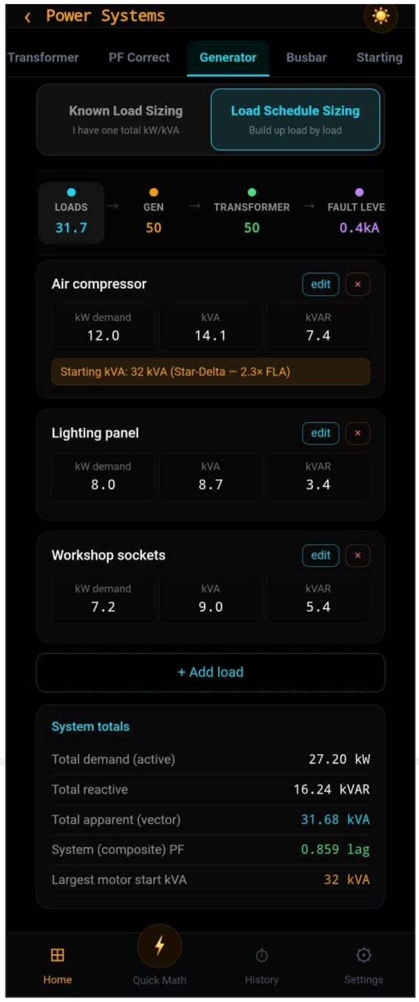


Figure — Power Systems: Generator multi-load sizing flow

Transformer, generator, and power-factor calculations at the system level.

Sub-tab	What it does
Transformer	Transformer parameters and fault current contribution.
PF Correct	Power factor correction capacitor sizing.
Generator	The full multi-load generator sizing tool, working through four stages in one flow: Loads (build up a load schedule, or use a single known-load shortcut), Gen (standard generator size, with altitude and ambient-temperature derating per ISO 8528-1), Transformer (matching step-up/step-down transformer sizing), and Fault Level (an IEC 60909 fault level calculation, for protection coordination). Use this for any generator sizing job, including multiple simultaneous loads or when you need fault-level figures downstream.
Busbar	Busbar current rating.
Starting	Motor starting comparison (DOL vs. reduced-voltage methods).

10. Power Quality

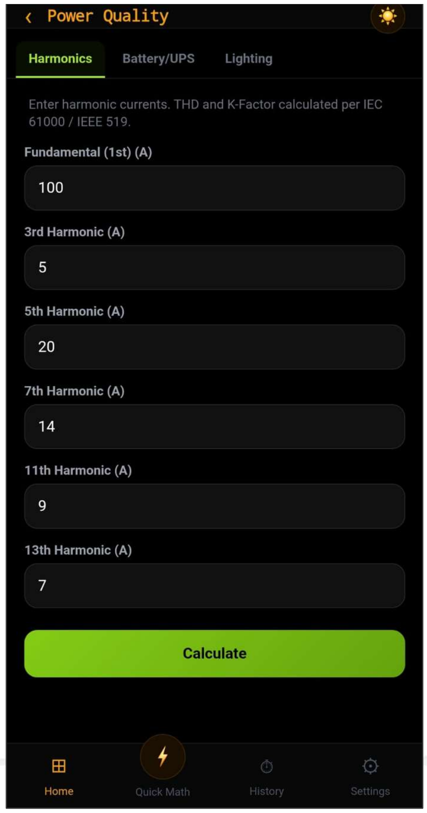


Figure — Power Quality: Harmonics (THD / K-Factor)

Harmonics, backup power, and lighting design.

Sub-tab	What it does
Harmonics	THD and K-factor calculations.
Battery/UPS	Battery and UPS autonomy sizing.
Lighting	Lumen-method lighting design per SANS 10114.

11. Renewable Energy

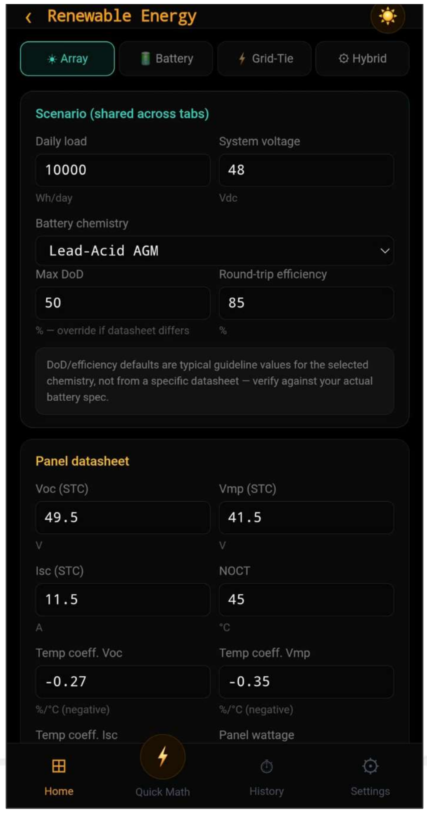


Figure — Renewable Energy: PV Array sizing

PV array sizing, battery sizing for off-grid and hybrid systems, grid-tie compliance for South African small-scale embedded generation (SSEG), and combined solar/battery/generator system design.

Sub-tab	What it does
Array	PV array sizing — works out the maximum and minimum number of panels you can safely put in series (checking cold-weather Voc against your inverter's maximum DC input, and hot-weather Vmp against the MPPT minimum voltage), how many strings can run in parallel on your controller's input rating, and the array size (in Wp and approximate panel count) needed to meet a daily energy target at your site's peak sun hours. Clearly warns if a panel/inverter pairing is incompatible rather than returning a silently wrong answer.
Battery	Battery bank sizing for off-grid or backup use — usable energy, required raw capacity (accounting for depth-of-discharge and round-trip efficiency losses), discharge-current check against the bank's rating, charge-controller sizing with a safety margin, and an MPPT-vs-PWM recommendation based on the ratio between array and bank voltage. Supports lead-acid (flooded/AGM/gel) and LiFePO4 chemistries.
Grid-Tie	NRS 097-2 categorization for grid-tied small-scale embedded generation (SSEG) systems in South Africa — classifies your system by capacity (Category A1/A2/A3, or flags it as above SSEG scope entirely) and works through a DC:AC ratio check against your inverter. Includes a structured compliance checklist covering utility registration, metering, anti-islanding protection, utility interface characteristics, phase balance, and fault-level/equipment rating requirements, each cited to the relevant clause.
Hybrid	Combined solar + battery + generator system design — battery bridge sizing for a low-sun design scenario, generator energy deficit and run-hours for an extended low-sun period, and a consistency check between your bridge days and extended

Sub-tab	What it does
	design period. Can pull a generator sizing result through directly from the Power Systems Generator tab if you've already sized one for this site.

Note: The Grid-Tie compliance checklist is written for South African SSEG requirements (NRS 097-2). If you are working in Lesotho, confirm the equivalent requirements with your utility directly before relying on this checklist as-is — it flags this jurisdiction gap explicitly rather than assuming South African rules apply unchanged.

12. Installation Design

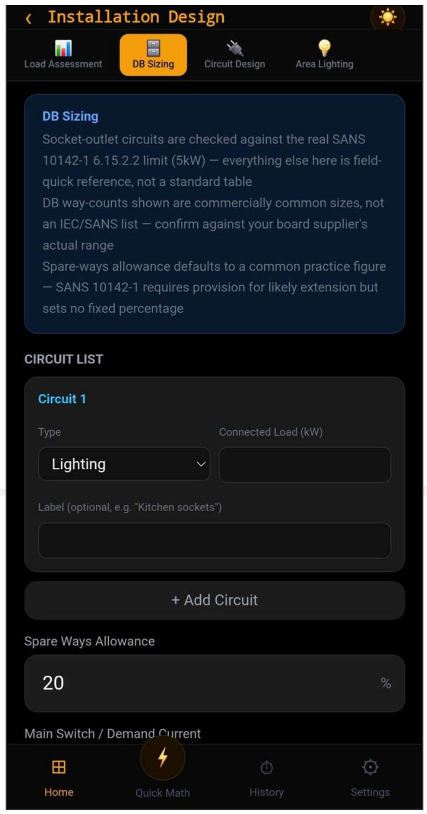


Figure — Installation Design: DB Sizing

Building and installation electrical design — from overall load assessment through to individual final-circuit sizing and exterior lighting layout.

Sub-tab	What it does
Load Assessment	Builds up a total connected load and maximum demand from a list of load categories, each with its own demand factor (which you set, since neither SANS 10142-1 nor its parent IEC 60364-1 mandates a fixed diversity table for this — both explicitly leave it to a registered person's judgement). Gives total connected load, total maximum demand, demand current, and a recommended main switch/breaker size.
DB Sizing	Distribution board sizing from a list of final circuits — flags any socket-outlet circuit exceeding the 5 kW limit, applies your chosen spare-ways allowance (always rounding up), and recommends both a standard DB size and (optionally) a main switch rating.
Circuit Design	Sizes an individual final circuit end-to-end: design current from the connected load, the next standard breaker rating above that current, and a cable size that satisfies both the breaker rating (not just the raw design current) and your voltage-drop limit for the given installation method and length.

Sub-tab	What it does
Area Lighting	Exterior/area lighting design using the lumen method — required lux level, fitting count, mounting height (always four times the pole spacing, an inherent relationship of the geometry used), and power density. Cross-references a lux guide across several risk-tier categories; four of those categories are cross-validated against ISO/CIE 8995-3, the rest (building sites, parking lots) are single-source and flagged as such.

Note: The Area Lighting pole-spacing/mounting-height relationship is a general lighting-geometry convention, not a SANS 10389-1 citation — SANS 10389-1's own content on this point was not directly accessible during development, and the tab says so explicitly rather than implying a false citation.

13. Overhead Reticulation

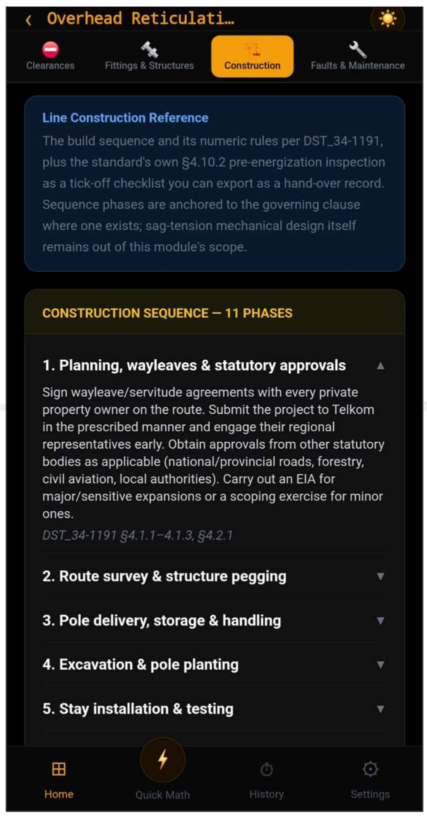


Figure — Overhead Reticulation: Construction sequence

Overhead-line conductor selection, spacing, clearances, fittings, construction reference, and fault-finding for medium-voltage bare-conductor reticulation. This module explicitly does not perform sag-tension mechanical calculations (wind/ice loading, span mechanics, pole strength design) — that remains genuine standalone structural-engineering scope, kept out of this field-quick tool by design.

Sub-tab	What it does
Conductor Sizing	Bare overhead conductor lookup — 47 conductors from Squirrel through IEC 800 (835mm²), covering ACSR, AAAC, and AAC types. 40 are fully rated with Rate A (normal/continuous) and Rate B (emergency) current ratings at four operating temperatures (50/60/70/80°C); 7 older BS215-family conductors (including Rabbit, still common on-site) are shown with real dimensional data but an honest no-ampacity note, since they are not covered by the current rating source. Bare conductors carry no insulation, so none of the Cable module's derating factors (grouping, ambient, installation method) apply here.

Sub-tab	What it does
Pole Spacing	Electrical span/phase-spacing calculation relating conductor swing under wind to the required phase spacing at the pole top for a given span. Verified for 22 kV only — other voltages are honestly flagged rather than extrapolated.
Pole Planting	Standard planting depths for wood and concrete poles (including transformer-pole variants), from the standard's own planting-depth table. Depths are picked from a fixed list of real stocked pole sizes — depths for a length not in that list are deliberately not interpolated or guessed.
Clearances	Minimum clearances (safety clearance, ground, above roads/railways, to buildings/vegetation, to communication/other power lines, and horizontal spacing) by voltage band, sourced directly from SANS 10280-1 Annex E (Table E.1) — a single verified table covering the full range from LV through 765 kV AC / 533 kV DC. Quick-select buttons are provided for the standard Eskom/national transmission voltage classes (132/220/275/400/765 kV), and a separate conductor-type selector (Bare/ABC/Concentric) appears for LV voltages, since ground clearance there depends on conductor type. Every clearance figure shown is a real, verified value from the standard — none of this table is an estimate or a placeholder.
Fittings & Structures	Preformed fitting selection (dead-ends, splices, armor rods, guy grips) matched by conductor diameter — the criterion every manufacturer catalogue actually uses — with an explicit warning that colour codes on fittings are manufacturer-specific and not standardized. Includes a collapsible qualitative reference on support structure types (suspension, strain, angle, terminal, transposition) and materials (wood, concrete, lattice steel, steel monopole).
Construction	Three parts in one tab: (1) an 11-phase line construction sequence from wayleaves through energization, each phase referenced to its governing clause; (2) a set of clause-cited stringing and construction numeric rules (tension limits, joint placement rules, stay assembly rating, minimum span, tensioning C-values); and (3) an interactive pre-energization inspection checklist reproducing the standard's own visual-inspection list, which you tick off item-by-item and export as a PDF hand-over record — any unticked item is shown clearly as incomplete in the exported record, since every item must be affirmative before a line may be energized.
Faults & Maintenance	A lightning-exposure calculator estimating expected strikes per year to a line from its geometry and your local ground flash density (or annual thunder days, if that's what you have available) — no default ground flash density is offered, since it varies by more than two orders of magnitude across the region and a real local figure matters. Also includes a qualitative fault-finding reference (common MV overhead fault mechanisms with what to look for on patrol) and a stringing-equipment glossary of trade terminology.

Note: Sag & tension, wind/ice loading, and pole strength/selection are deliberately out of scope for this module — that is genuine structural-engineering design territory, not a field-quick calculation, and would need its own dedicated tool if ever added.

Note: Ground flash density (Ng) for the lightning-exposure calculator must be a real figure for your area, from your utility's isokeraunic data or Weather Bureau records. The app will not suggest a default value, since a wrong Ng can make the estimate meaningless.

14. Unit Converter

Twelve conversion categories in one place: Power, Energy, Current, Voltage, Resistance, Length, Area (Cable), Pressure, Torque, Temperature, Mass, and Speed. Select a category, then convert between any two units within it.

15. Formula Reference

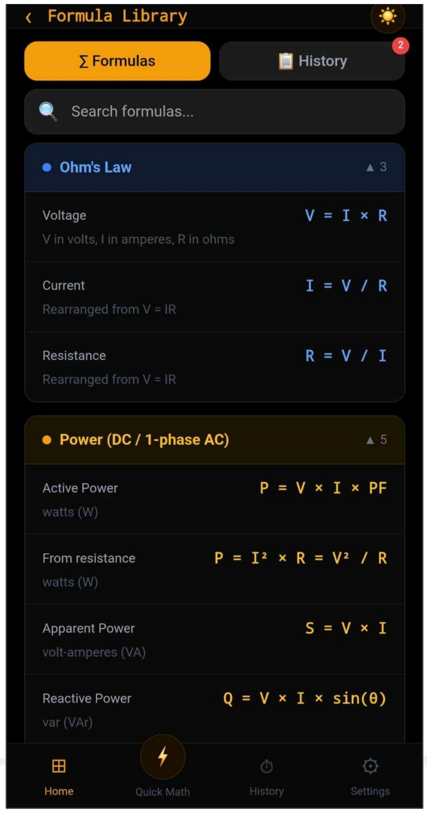


Figure — Formula Library

A searchable library of the formulas used throughout the app, organized by category, plus a Useful Constants section (mathematical constants, copper/aluminium resistivity, temperature coefficients, and physical constants like permittivity and permeability of free space). Also shows your recent calculation history inline.

16. Quick Math

A standard calculator plus a library of your own saved formulas — accessible from the lightning-bolt icon in the bottom navigation, on top of whatever module you're currently in.

16.1 Calculator

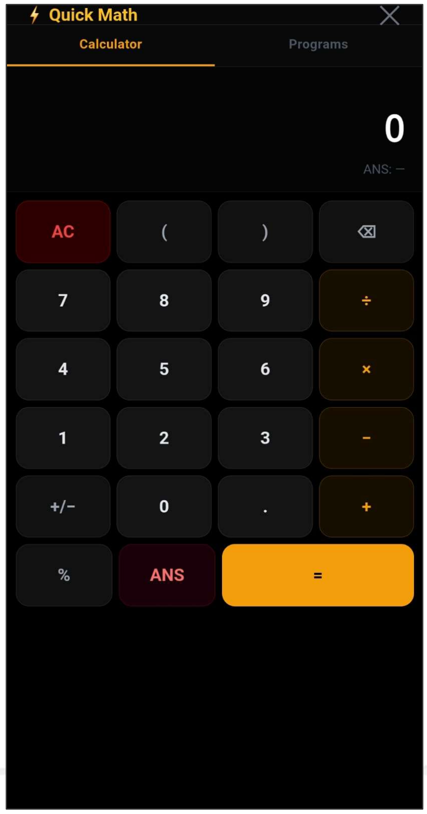


Figure — Quick Math: standard calculator

A standard four-function calculator: digits, parentheses, percent, and the four basic operators, plus +/-, backspace, clear, and ANS (recalls your last result). It does not include scientific functions (trig, log, etc.) — for those, use the relevant calculation module directly.

16.2 Programs — your own saved formulas and worksheets

The Programs tab lets you save your own formulas and reuse them without retyping. Tap "+ New", give it a name and description, and write the formula using standard operators plus √ for square root and functions like sin, cos, tan, acos, sqrt, log, ln. PowerSuite automatically detects which parts of your formula are fill-in values (variables) versus built-in functions, and lets you give each variable a friendlier name for the fill-in prompt.

You can also build multi-step worksheets — a formula whose result feeds into a second formula, and so on — with an optional Pass/Fail check at the end (e.g. "is % voltage drop under 5?"). Four examples come pre-loaded:

- kVA→A — three-phase kVA to amps
- Volt Drop — single-phase voltage drop in millivolts
- PF Correct — capacitor kVAr needed for power factor correction
- 3Ø Feeder Check — a worksheet: full-load current, then % voltage drop, then a Pass/Fail check against a limit you set

Note: Any Pass/Fail limit you set (e.g. a 5% voltage drop threshold) is something you configure yourself — it is not a fixed rule built into the app. Confirm the correct limit for your installation against SANS 10142 or your project specification before relying on it.

17. Exporting Results as PDF

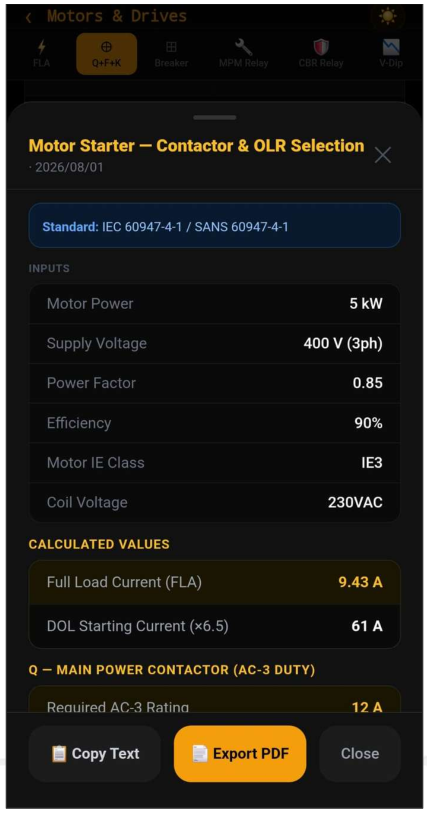


Figure — PDF export dialog (Motor Starter result)

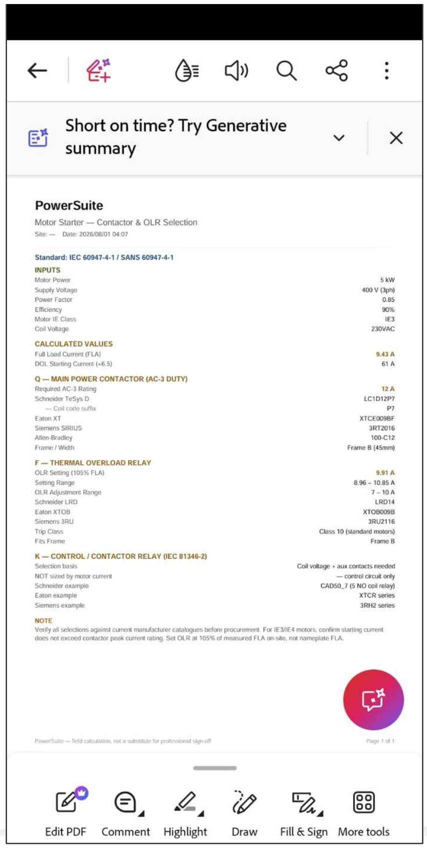


Figure — The resulting exported PDF

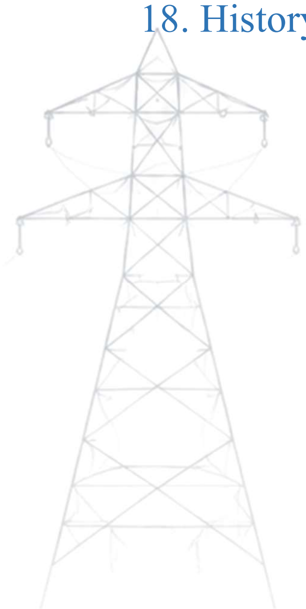
Every calculation module can generate a Result Card — a clean summary of your inputs and results, with the relevant standard shown where applicable. From a Result Card you can either:

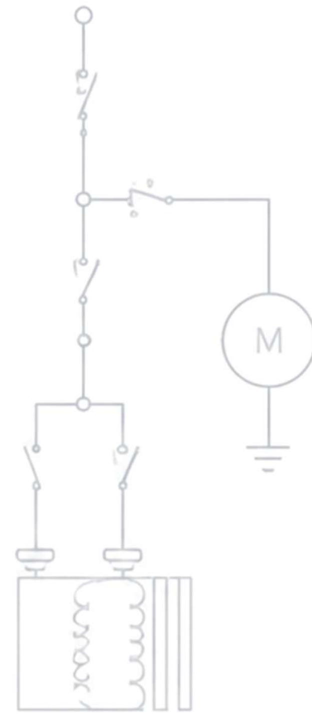
- Share Text — copies a plain-text summary to your clipboard or opens your phone's share sheet, quick for a WhatsApp message.
- Export PDF — generates a formal PDF document with your inputs, results, and any notes, which you can save, email, or share like any other file on your phone.

When you tap Export PDF, you'll be shown a suggested filename (built from the calculator name, site, and timestamp) which you can edit before generating the file. Once created, your phone's normal share screen opens so you can send or save it wherever you need.

Note: The Overhead Reticulation Construction tab's pre-energization checklist exports slightly differently: it is a hand-over compliance record, not a single calculation. Unticked items are shown clearly as incomplete in the exported PDF, and the export states plainly whether every item was affirmative at the time of export.

18. History





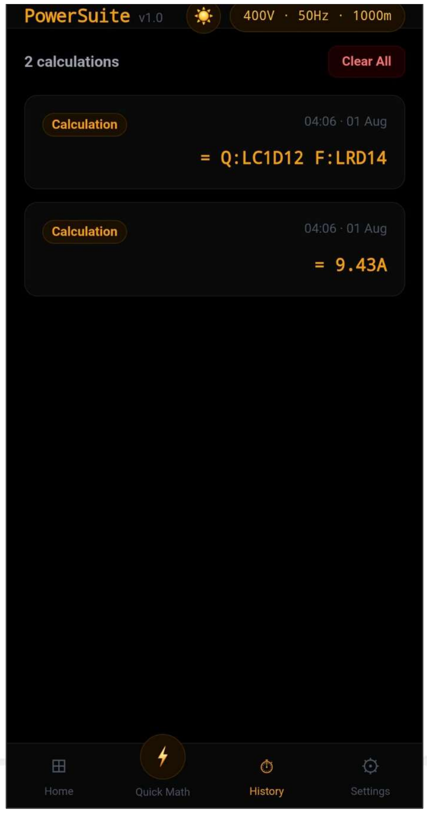


Figure — History: saved calculations

A running log of your recent calculations across every module, most recent first, so you can retrace what you worked out earlier in the day without redoing it.

19. Settings

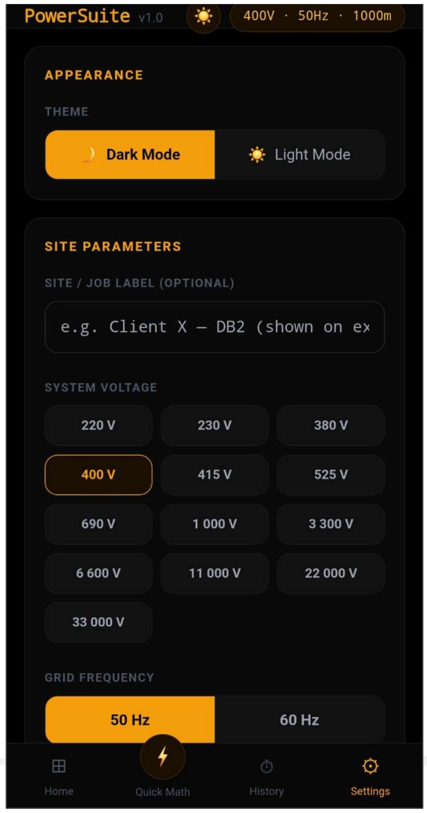


Figure — Settings: Site Parameters

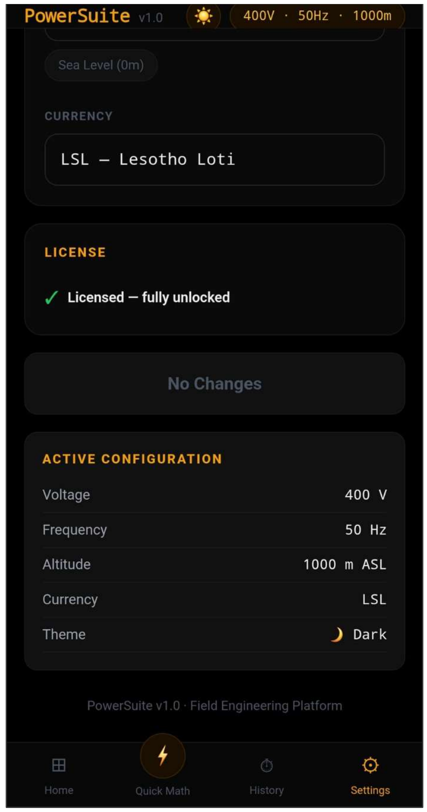


Figure — Settings: License status and Active Configuration

19.1 Site Profile

Sets the defaults used across calculators: system frequency, site altitude, and currency. Setting this once per site saves re-entering the same values in every module.

19.2 Appearance

Toggle between dark and light theme using the sun/moon icon in the header.

19.3 License

Shows your trial status and days remaining. If you already have a license key, you can activate it here immediately — you do not need to wait for the trial to expire.

20. Licensing

20.1 How it works

PowerSuite uses a 90-day free trial per device, followed by a once-off license key purchased from your supplier. The trial clock and license status are both tracked on a remote server, not on your phone, so the trial cannot be reset by clearing app data or reinstalling.

Outside of that one-time trial registration and periodic license checks, PowerSuite never requires connectivity — every calculation, on every module, works with zero signal.

20.2 Activating a license key

You'll receive a key in the format HETSA-XXXX-XXXX-XXXX. Enter it either at the trial-expiry screen, or ahead of time via Settings → License. Each key is valid for one device at a time.

20.3 "This license key is already activated on another device"

This means the key is currently bound to a different device (or the same physical phone under a different install identity — for example, after certain app updates). This is a safeguard, not a bug. If you believe this is your own key and device, contact your supplier to have it released for re-activation.

21. Troubleshooting

Sub-tab	What it does
"Checking license…" takes a while	Normal on first use if the server hasn't been contacted recently — it can take up to a minute to respond. Just wait; it isn't stuck.
"Couldn't reach the license server"	Check your internet connection is actually working (try loading any website), then tap Retry.
"Already activated on another device"	See section 20.3 above — contact your supplier if this is genuinely your key and device.
A field won't accept a decimal point	Use a period (.) — a comma will be automatically converted for you.
Export PDF doesn't seem to do anything	Check the filename field isn't empty, then check your phone's notification shade — the share screen sometimes opens behind the app briefly.
The Overhead Reticulation Clearances tab shows a message instead of a number for a voltage above 765 kV AC / 533 kV DC	This is expected, not a bug. This is expected, not a bug. Verified clearance figures cover the full SANS 10280-1 Annex E, Table E.1 range — LV through 765 kV AC / 533 kV DC. A voltage entered outside that range is intentionally flagged rather than showing a guessed number.

22. Feature Maturity

Everything documented in this manual is available now — fully built and verified, not a beta or preview feature. There is no separate maturity tier in this release; if a module or sub-tab is described in this manual, it works as described.

One specific technical scope limitation is worth restating here for visibility, since it is already flagged on the relevant tab itself:

Note: Cable & Wiring's Duct Derating (part of Underground Reticulation) reuses the direct-buried grouping-derating table rather than a duct-bank-specific one, since a duct-specific grouping table could not be verified against IEC 60364-5-52 with confidence at the time of writing. Treat it as a reasonable, conservative estimate for typical field-quick sizing — not a substitute for a full thermal study on a large duct-bank installation. This is the only known scope limitation across the app; every other feature works exactly as documented.

23. Support

For license issues, feature requests, or anything not covered here, contact your PowerSuite supplier.

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